TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Scalable manufacturing and qualification of sulfide solid electrolytes for all-solid-state batteries, enabled by TNO materials metrology, interface engineering, and physics-based digital twins			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Battery Materials	Batteries	Manufacturing and Measurement and Evaluation Equipment
Concept of Proposed Technology	Core concept: Jointly develop an industrialisation-ready framework for sulfide solid electrolyte (SSE) production and qualification. Approach: Combine the Korean SME's scalable wet synthesis route with TNO's independent metrology, interface engineering, and physics-based modelling. Key elements:			

Rationale for Joint R&D Project	• Materials metrology and quality gates: phase purity and composition control, particle size distribution, pellet or separator fabrication, and ionic transport characterisation using multi-current EIS and ageing facilities. [3][2][1] • Moisture and safety-by-design: quantify degradation and by-product formation under controlled humidity and temperature, and translate results into process and handling requirements. [15][4] • Interface engineering: apply scalable spatial atomic layer deposition (sALD) and dry coating know-how to design artificial solid-electrolyte interfaces that mitigate interfacial resistance and ageing. [4][6] • Digital twin and BPX-ready parameterisation: build a process-structure-property digital twin that links synthesis parameters to electrochemical performance and delivers physics-based parameter sets for downstream cell and pack design. [8][10][11][12]
	Sulfide SSEs enable high room-temperature ionic conductivity, but practical deployment is limited by moisture sensitivity, interface instability, and scale-up variability. [15][16] The Korean partner contributes a scalable wet-chemical synthesis pathway. TNO contributes independent test and validation infrastructure, scalable interface and manufacturing technologies from Holst Centre, and physics-based modelling, diagnostics and digitalisation methods (including Battery Passport readiness and BPX workflows). [4][1][13][12] Holst Centre reports dry coating can cut manufacturing costs by up to 20% and energy usage by up to 50%, and eliminates toxic solvents such as NMP used in traditional electrode production. [4] Together, this reduces time-to-market by turning lab-scale materials into repeatable, spec-driven batches and validated prototype cells suitable for industrial evaluation.
Key Words	1. Sulfide solid electrolyte (SSE) [15] 2. Liquid-phase or wet-chemical synthesis [16] 3. Moisture stability and H2S mitigation [15][6] 4. Interface engineering with sALD and dry coating [4][6] 5. Physics-based digital twin and BPX parameter exchange [8][11][12]

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	1. Define critical-to-quality (CTQ) metrics and test protocols for sulfide SSE batches: chemistry, phase purity, particle morphology, and ionic transport. [15][2] 2. Establish a scalable synthesis and post-processing window (mixing, solvent handling, drying, milling, densification) using design-of-experiments (DoE) linked to CTQs. [16] 3. Develop scalable interface engineering (coatings and dry-coated composite layers) to reduce interfacial impedance and improve cycling stability in representative ASSB cells. [4][6][15] 4. Build a physics-based digital twin that maps CTQs to cell-level performance, delivering BPX-ready parameter sets for downstream design and control studies. [8][11][12] 5. Demonstrate prototype cell performance using partner-provided materials and TNO pre-pilot manufacturing capability for coin and pouch cells. [4][5]			
R&D Target Specifications	Target specifications (to be finalized jointly based on partner material class and application): • Room-temperature ionic conductivity of SSE separator or pellet: ≥1 mS/cm at 25°C (baseline threshold; higher values targeted where feasible). [4][15] • Batch-to-batch variability: CTQ metrics within agreed statistical limits (Cp/Cpk targets defined in WP1). • Moisture stability: reduced conductivity loss and reduced H2S generation versus baseline under controlled humidity exposure tests (protocol derived from literature and safety requirements). [15][4] • Interfacial resistance: demonstrable reduction versus uncoated baseline through artificial SEI coatings or engineered composites (validated by EIS). [15][4] • Prototype demonstrator: coin cell and/or pouch cell prototypes assembled on Holst Centre pre-pilot line for industrial compatibility assessment. [4][5]			
R&D Plan	Duration: 36 months (indicative). WP1 (M1-M3): Requirements and CTQ definition. Select representative cell formats and operating envelopes; define metrology plan and data model. [15][2] WP2 (M1-M12): Baseline materials and process mapping. Sample exchange, wet synthesis parameter screening, post-processing study (drying, milling, densification), initial conductivity and stability screening. [16][15] WP3 (M3-M24): Advanced characterisation and ageing. Multi-channel testing and high-current EIS (up to 128 A) to quantify transport and degradation drivers, plus controlled temperature and humidity exposure. [3][1] WP4 (M6-M30): Interface and layer engineering. Apply sALD or dry-coating derived concepts to form artificial interfaces, composite catholytes, or separator layers; evaluate interfacial impedance and cycle			

	stability. [4][6] WP5 (M6-M30): Digital twin and BPX-ready parameterisation. Parameter identification from EIS and cycling data; physics-based models and sensitivity analyses; export parameter sets using BPX. [8][10][11][12] WP6 (M18-M36): Prototype build and industrial assessment. Coin and pouch cell prototypes on Holst Centre pre-pilot line; pilot batch acceptance tests; techno-economic and scale-up recommendations; optional Battery Passport data fields for EU readiness. [4][13][14] Milestones: (i) CTQ and test protocol agreed (M3), (ii) stable process window demonstrated on lab batches (M12), (iii) interface-engineered cells validated (M24), (iv) BPX-ready digital twin released (M30), (v) pilot prototypes and final guidelines delivered (M36).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) responsibilities: • Project leadership support, test plan definition, and independent validation of partner-produced SSE batches. [2][7] • Battery Lab electrochemical test infrastructure including: independent cell test channels up to 100 A (parallelable up to 1200 A), pack test capability up to 750 V and 400 A (continuous, bi-directional, regenerative), and controlled climate environments with adjustable humidity and safety provisions. [3] • Advanced Electrochemical Impedance Spectroscopy (EIS) capability for cells up to 128 A and 10 kHz (lower-current EIS up to 100 kHz), plus multi-channel ageing characterisation and hardware-in-the-loop capable setups for algorithm and model validation. [3][1] • Solid-state and manufacturing expertise via Holst Centre: dry coating and thin-thick film deposition processes, pre-pilot line for coin and pouch cell assembly, and scalable sALD for artificial solid-electrolyte interfaces. [4][6] • Physics-based modelling and digital twins at TNO Powertrains, including BPX-ready parameterisation workflows and model-based battery management concepts. [7][8][10][11][12] • Optional digital thread for regulatory readiness: mapping of process and performance data to Battery Passport-relevant fields for EU market access from February 2027. [13][14]
Expected R&R of Korean SME Partner	Korean SME Partner responsibilities: • Provide sulfide SSE materials produced via the partner's scalable wet-chemical synthesis route, including baseline and improved batches and relevant process metadata. • Support joint DoE by varying synthesis and post-processing parameters within agreed safety and IP boundaries; deliver pilot-scale batches for acceptance testing. • Provide application requirements and constraints (target cell format, operating envelope, manufacturability, and cost targets), and support prototype build inputs (powder handling, densification approach, layer thickness targets). • Participate in joint reviews, share non-confidential data needed for model calibration, and evaluate results for technology transfer and potential commercialisation.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical outcomes: • Repeatable sulfide SSE product specification with CTQ metrics and validated test methods (conductivity, stability, and interface impedance). [15][2] • Documented process window and scale-up guidance for wet synthesis and post-processing, with quantified sensitivity of key parameters. [16] • Engineered interfaces and layer processing concepts using scalable sALD and dry coating, demonstrated in representative ASSB cell builds. [4][6][15] • A process-structure-property digital twin delivering BPX-ready parameter sets and uncertainty bounds to accelerate downstream cell, pack, and BMS engineering. [8][10][11][12] Commercial and societal impact: • De-risking of scale-up and faster qualification cycles for next-generation, safer batteries relevant to EU markets and heavy-duty electrification. [4][7] • Improved readiness for upcoming EU Battery Passport requirements (mandatory from 18 February 2027 for specified battery categories), via early alignment of data capture and traceability. [13][14]

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization

	Commercialization Stage (9)	9th Stage: Commercialization